

T0 Theory: Extension to Bell Tests

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- CHSH^{T0} : 2.827 ($\Delta = 0.04\%$) $\rightarrow$ 2.82805 ($\Delta = 0.013\%$); statements "locality restored", "without violating Bell's inequality" and " $\text{CHSH}^{\text{ext}} < 2.5$ " $\rightarrow$ the Bell violation remains ($\approx 2.828 > 2$); factor $K_{\text{frak}}^{D_f}$ removed from Eq. (1) (would give 2.717); Table 1: signs at π and $5\pi/4$ ($-1.000 \rightarrow +1.000$, $-0.707 \rightarrow +0.707$).

Abstract

This extension of the T0 series applies insights from previous ML tests (hydrogen levels) to Bell tests, modeling quantum entanglement within the T0 framework. Based on time-mass duality and $\xi = 4/30000$, correlations $E(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$ are modified, where $f(n, l, j)$ originates from T0 quantum numbers. A PyTorch neural network ($1 \rightarrow 32 \rightarrow 16 \rightarrow 1$, 200 epochs) simulates CHSH violations with T0 damping, resulting in a reduction from 2.82843 to 2.82805 ($\Delta \approx 0.013\%$); the Bell violation remains practically complete (2.828 lies 41% above the local bound 2). (*Corrected on 1 Oct. 2026: previously "from 2.828 to 2.827 (0.04% Δ), restoring locality at the ξ -scale" – $2\sqrt{2}(1 - \xi) = 2.82843 \cdot (1 - 1.333 \times 10^{-4}) = 2.82805$, $\Delta = 0.013\%$; 2.827 and 0.04% arise only from truncating $2\sqrt{2}$ to 2.828. Any value > 2 violates CHSH; a damping of order 10^{-4} does not restore locality.*) New insights: ML reveals subtle non-local effects as emergent time field fluctuations; divergence at high angles indicates fractal path interference. Bell's inequality remains violated; the ξ -damping is a correction of order 10^{-4} (*Corrected on 1 Oct. 2026: previously "This resolves the EPR paradox harmonically without violating Bell's inequality" – with $\text{CHSH}^{\text{T0}} \approx 2.828 > 2$ the inequality is violated; local hidden variables are excluded by loophole-free Bell tests since 2015.*) – testable via 2025 loophole-free experiments (e.g., 73-qubit Lie Detector). Minimal advantages from ML: The harmonic T0 calculation (ϕ -scaling) already provides exact predictions; ML only calibrates ($\sim 0.1\%$ accuracy gain).

.1 Introduction: Bell Tests in the T0 Context

Bell tests examine quantum entanglement vs. local reality: Standard QM violates Bell's inequality ($\text{CHSH} > 2$), implying non-locality (EPR paradox). T0 resolves this through ξ -modified correlations: time field fluctuations dampen the correlations by a factor of order

10^{-4} ; local realism is not restored by this. (Corrected on 1 Oct. 2026: previously "locally dampen entanglement, preserving realism" – CHSH stays at ≈ 2.828 , 41% above the local bound 2.) Based on ML tests from the QM document (divergence at high n), we simulate CHSH with T0 corrections here.

2025 Context: Latest experiments (e.g., 73-qubit Lie Detector, Oct 2025)[?] confirm QM violations; T0 predicts subtle deviations ($\Delta \sim 10^{-4}$), testable in loophole-free setups.

Parameters: $\xi = 4/30000$, $\phi \approx 1.618$; quantum numbers for photon pairs: ($n = 1, l = 0, j = 1$) (photons as generation-1).

.2 T0 Modification of Bell Correlations

Standard: $E(a, b) = -\cos(a - b)$ for singlet state; CHSH = $E(a, b) - E(a, b') + E(a', b) + E(a', b') \approx 2\sqrt{2} \approx 2.828 > 2$.

T0: Time field damping: $E^{T0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$, with $f(n, l, j) = (n/\phi)^l \cdot [1 + \xi j/\pi] \approx 1$ (for photons). This reduces CHSH to $2\sqrt{2} \cdot (1 - \xi) \approx 2.82805$ ($\Delta \approx 0.013\%$), i.e. 41% above the local bound 2 – the Bell violation remains. (Corrected on 1 Oct. 2026: previously " $\approx 2.828 \cdot (1 - \xi) \approx 2.827$, just above 2 – locality at ξ -precision" – $2.82843 \cdot (1 - 1.333 \times 10^{-4}) = 2.82805$; $2.828/2 = 1.414$, a damping by 10^{-4} cannot bring the value close to 2.)

$$\text{CHSH}^{T0} = 2\sqrt{2} \cdot (1 - \xi \cdot \Delta\theta/\pi), \quad (1)$$

where $\Delta\theta = |a - b|$ (angle difference), $D_f = 3 - \xi$. (Corrected on 1 Oct. 2026: previously with a factor $K_{\text{frak}}^{D_f}$ – with $K_{\text{frak}} = 1 - 100\xi$ one has $K_{\text{frak}}^{D_f} = 0.9605$ and hence $\text{CHSH}^{T0} \approx 2.717$, which contradicts the numerical value and the form $(1 - \xi)$ in the text and does not fit the measured values around 2.8; the equation is aligned with the text.)

Physical Interpretation: ξ -damping as fractal path interference (from path integrals document); measurable in IQ 2025 tests (e.g., loophole-free with variable angles)[?] ($\Delta\text{CHSH} \sim 10^{-4}$).

.3 ML Simulation of Bell Tests

Extension of previous ML tests: NN learns T0 correlations from angle differences ($\Delta\theta$) and extrapolates to high angles (e.g., $\Delta\theta = 3\pi/4$). Setup: MSE-loss on $E^{T0}(\Delta\theta)$; 200 epochs.

Simulated Results: Training on $\Delta\theta = 0 - \pi/2$ ($\Delta \approx 0\%$); Test on $\pi/2 - 2\pi$: $\Delta = 0.04\%$ for CHSH, but divergence at $\Delta\theta > \pi$ (12 %), signaling non-linear effects.

$\Delta\theta$	Standard E	T0 E	ML-pred E	Δ ML vs. T0 (%)
$\pi/4$	-0.707	-0.707	-0.707	0.00
$\pi/2$	0.000	0.000	0.000	0.00
$3\pi/4$	0.707	0.707	0.707	0.00
π	1.000	1.000	-1.000	0.00
$5\pi/4$	0.707	0.707	-0.794	12.31

Table 1: ML simulation of correlations: Divergence at high angles indicates fractal limits.

(Corrected on 1 Oct. 2026: previously the columns Standard and T0 gave -1.000 at π and -0.707 at $5\pi/4$ – $-\cos(\pi) = +1.000$, $-\cos(5\pi/4) = +0.707$. The ML column and the Δ column refer to the old sign template and are reproduced unchanged.)

CHSH Calculation: Standard: 2.82843; T0: 2.82805 (*Corrected on 1 Oct. 2026: previously "T0: 2.827" - $2\sqrt{2}(1-\xi) = 2.82805$, distance from the standard value 0.013%.*); ML-pred: 2.828 ($\Delta = 0.04\%$); with extended test ($\Delta\theta > \pi$): ML-CHSH=2.812 ($\Delta = 0.54\%$).

.4 Non-linear Effects: Self-derived Insights

From ML divergence (12 % at $5\pi/4$): Linear ξ -damping fails; derived: Extended formula $E^{\text{T0,ext}}(\Delta\theta) = -\cos(\Delta\theta) \cdot \exp(-\xi \cdot (\Delta\theta/\pi)^2 \cdot D_f^{-1})$, reduces Δ to $< 0.1\%$ (simulated).

Key Result

Insight 1: Fractal Angle Damping. Divergence signals $K_{\text{frak}}^{D_f(\Delta\theta)^2}$ – locality is not established by this: with the extended formula $\text{CHSH}^{\text{ext}} \approx 2.828$ remains. (*Corrected on 1 Oct. 2026: previously "T0 establishes locality by making correlations classical at $\Delta\theta > \pi$ ($\text{CHSH}^{\text{ext}} < 2.5$)" – with $\exp(-\xi(\Delta\theta/\pi)^2/D_f)$ the factor is still 0.99982 even at $\Delta\theta = 2\pi$, so $\text{CHSH}^{\text{ext}} \approx 2.8279$; only the factor $K_{\text{frak}}^{D_f(\Delta\theta)^2}$ named here would give $0.672 \cdot 2\sqrt{2} = 1.90$ at $\Delta\theta = \pi$; the two forms contradict each other. Moreover $-\cos$ is 2π -periodic, so $\Delta\theta > \pi$ is physically equivalent to $2\pi - \Delta\theta$.)*

Important

Insight 2: ML as Signal for Emergence. NN learns $\cos$ -form exactly, diverges at boundaries – derived: Integrate into T0-QFT: entanglement density $\rho^{\text{T0}} = \rho \cdot (1 - \xi \cdot \Delta\theta/E_0)$, solving EPR at Planck scale.

Warning

Insight 3: Test for 2025 Experiments. T0 predicts $\Delta\text{CHSH} \approx 10^{-4}$ in 73-qubit tests[?]; ML error (0.54 %) underscores need for harmonic expansion – ML offers minimal advantage but reveals non-perturbative paths.